MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5691

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, medium Mu

Outline	---	9-39 (EIA)
Base	---	38-58, phenolic
Envelope	---	T9
Catnode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2j	2a	2k	1g	1a	1k	n	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rg	Ik/k	Ic/c	Po/a	TA	Alt
Unit:	V	Vdc	Vdc	v	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.6	275	-1 (see note 1)	100	0.25	10	2	1.0	90	Note 2
Minimum:	6.0	---	-100	---	---	---	---	---	-55	---
TEST CONDITIONS:	6.3	250	-2	---	---	---	---	---	---	---

GENERAL:

(D) Qualification - Not required

(D) denotes changes

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1266	Total grid current	3,4	Rg = 0.5 Meg	0.65	II	Ic	0	-0.3	µAdc
1256	Electrode current (1) (anode)	5		0.65	II	Ib	1.7	2.9	mAdc
1256	Electrode current (2) (anode)	6		0.65	II	ΔIb	---	0.6	mAdc
1256	Electrode current (3) (anode)	5,7	Ec = -5.5 Vdc	0.65	II	Ib	---	15	µAdc
1321	AC amplification	5	Esig = 0.2 Vac; Ebb = 100 Vdc; Ecc = 0	0.65	II	Eo	7.8	---	Vac
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1306	Transconductance (1)	5		2.5	I	Gm	1,300	1,900	µmhos
1306	Transconductance (2)	5	Ef = 5.7 V	4.0	I	ΔGm Ef	---	15	%
1301	Heater current			---	---	If	580	620	mA
1336	Heater-cathode leakage	3		---	---	Ihk	---	7.5	µAdc
1211	Insulation of electrodes			4.0	---	R	300	---	Meg
---	Anode current unbalance	3		4.0	Code G	ΔEb	-10	+10	Vdc
1316	Amplification factor	5		---	---	Mu	60	80	---
1246	Audio frequency noise	3	Ebb = 225 Vdc; Rb = Rg = 0.25 Meg; Rk = 3,200 ohms; Esig = 12 mVac	2.5	I	En	---	17	vu
1031	Low-frequency vibration	3	Rb = 2,000 ohms	---	Code G	Eo	---	40	mVac
1331	Direct-interelectrode capacitance		Without shield	---	---	C1g-1p	3.1	4.1	pF
						C1in	1.9	2.9	pF
						C1out	1.8	2.8	pF
						C2g-2p	3.1	4.1	pF
						C2in	2.2	3.2	pF
						C2out	2.1	3.1	pF
						C1p-2p	0.29	0.39	pF

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 -Continued								
1041	Shock	3	450 G; Ehk = +100 Vdc	---	---	---	---	---	---
1031	Vibration-fatigue	9		---	---	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	Low-frequency vibration	3	Rp = 2,000 ohms	---	---	Ep	---	100.	mVac
1336	Heater-cathode leakage	3		---	---	Ihk	---	30	µAdc
1306	Transconductance(1)	5		---	---	Sm	1,200	2,000	µmhos
1105	Permanence of marking			---	---	---	---	---	---
	Quality conformance inspection, part 3								
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 100 Vdc; Eb = Ec = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	3		---	---	Ihk	---	15	µAdc
1501	Intermittent life	10	Group A; Ebb = 275 Vdc; Rk/k = 450 ohms; Rg = 1.0 Meg; Ehk = 100 Vdc	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):								
1321	AC amplification	5	Esig = 0.2 Vac; Ebb = 100 Vdc; Ecc = 0	---	---	Ep	7.6	---	Vac
1306	Transconductance(1)	5		---	---	Sm	1,250	---	µmhos
---	Intermittent life-test end point (1,000 hours):								
1306	Transconductance(1)	5		---	---	Sm	1,000	---	µmhos
(D)1216	Base material insulating quality	11	Zone 5 (minimum)	---	---	---	---	---	---
(D)1031	Variable-frequency vibration	3,11	Rp = 2,000 ohms	---	---	Ep	---	40	mVac

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NOTES:

1. Maximum grid bias of -0.5 volt permissible for resistance-coupled amplifier service.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. Test with units connected in parallel.
4. This test shall be performed at the conclusion of the holding period.
5. Test each unit separately.
6. Electrode current (2) is the difference in electrode current (1) between units.
7. Apply Eb = 250 Vdc and Ec = -2 Vdc to unit not under test.
8. Test in the circuit of figure 1. Read ΔEb on the vacuum-tube voltmeter, the input resistance of which shall be not less than 10 megohms. Test tube in upright position. Tap tube perpendicular and parallel to a plane through pins 4 and 8. The instantaneous change in the unbalance shall not be more than ±0.4 volt during tapping.
9. (D) This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
10. Connect anodes together and connect grids together; use separate cathode resistors in each unit.
11. (D) This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

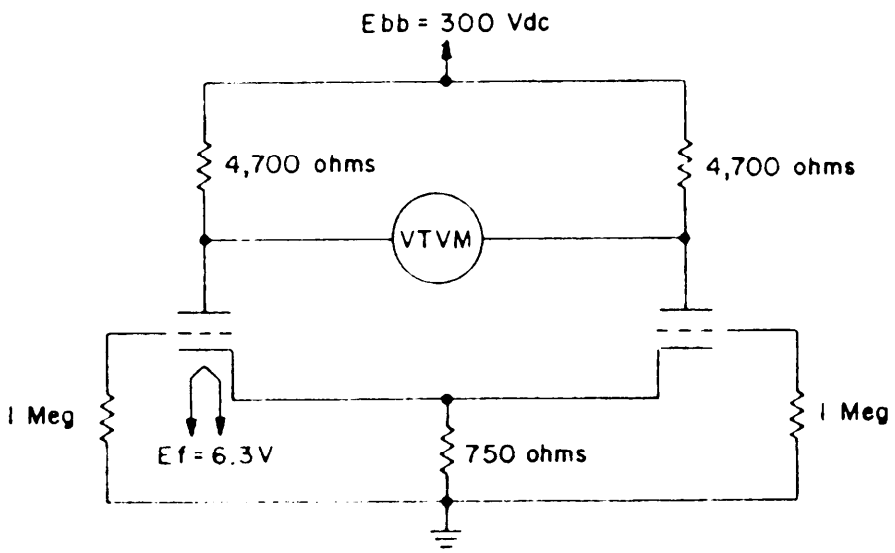


FIGURE 1. Circuit diagram.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 99
DLA - ES

User activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3354)

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